NEWTON SCHOOL OF TECHNOLOGY

Data Visualization & Analytics — Capstone Project | 2024 - 2028

SECTOR: FOOD & BEVERAGE RETAIL

Cafe Analytics Revenue Intelligence

Report

Item Performance, Payment Behavior & Temporal Sales Analysis

PROBLEM STATEMENT	DATASET	PERIOD
Revenue Drivers & Peak Sales	6,658 Validated Transactions	February 2026

GROUP 17 | SECTION A | Academic Year 2024 - 2028

SECTION 1

Executive Summary

This report presents a comprehensive analytical investigation into the revenue dynamics of a cafe retail operation, drawing on 6,658 validated point-of-sale transactions spanning January through December 2023. The central research question -- *how do item performance, payment preferences, and time-based sales trends impact overall cafe revenue, and which factors drive peak sales periods?* -- is answered systematically through quantitative evidence derived from a rigorously cleaned dataset.

Problem

Raw POS exports contained 33.42% corrupt, duplicate, or logically invalid records. Without structured data governance, business decisions would rest on a revenue base inflated by $2,847 in calculation errors and distorted by 87 duplicate transactions. The absence of reliable analytics prevented the identification of top-performing items, optimal payment infrastructure investment, and demand-aligned staffing strategies.

Approach

A seven-stage data engineering pipeline transformed 10,000 raw records into 6,658 production-grade transactions. Nine pivot-based analyses then decomposed revenue across four dimensions: product performance, payment method behaviour, temporal dynamics, and service channel (in-store vs. takeaway). All transformations were executed in Google Sheets and are fully reproducible.

Key Findings

- Total verified revenue: $60,483.50 across 6,658 transactions (ATV: $9.09).
- Salad is the dominant revenue generator at $15,963.50 (26.4% of total), nearly double the next-ranked item.
- Cash leads payment channel revenue at $23,649.00 (39.1%), though digital payments are closing the gap in Q4.
- March and January are peak revenue months ($5,274 and $5,290 respectively), while May records the lowest performance ($4,552.50).
- In-store transactions account for 70.3% of total revenue ($42,522), indicating the cafe's primary customer base is dine-in.
- The revenue distribution is relatively stable month-to-month (range: $4,552-$5,290), suggesting consistent demand with moderate seasonal variation.

Key Recommendations

- Invest in Salad supply chain reliability and introduce premium variants to expand its already-dominant revenue share.
- Launch a targeted digital wallet incentive programme in Q4 to capitalise on observed growth in non-cash transactions.
- Implement demand-aligned staffing in January, March, and June -- the three highest-revenue months.
- Deploy targeted promotions in May and February to address seasonal troughs and smooth revenue distribution.

SECTION 2

Sector & Business Context

The food service retail sector -- encompassing cafe operations, quick-service restaurants, and specialty beverage outlets -- operates on thin margins with high transaction velocity. Revenue optimisation in this environment depends critically on three levers: menu engineering (which items to promote), operational throughput (peak-period staffing), and payment infrastructure (which channels to prioritise).

Modern cafe operations generate rich transactional data through POS systems, yet a significant proportion of businesses fail to leverage this data effectively. Without systematic analytics, management teams lack visibility into which products drive disproportionate revenue, how customer payment preferences evolve, and when demand peaks occur -- all of which are foundational inputs to resource allocation and growth strategy.

Why This Problem Was Chosen

The cafe sector was selected because its transactional structure -- high frequency, multi-dimensional categorical variables, and clear revenue outcomes -- creates an ideal analytical environment. The problem statement directly addresses operational levers that management can act upon: menu composition, payment acceptance strategy, and demand-aligned scheduling. The dataset's temporal completeness (all 12 months of 2023) enables reliable seasonal analysis, and the presence of location data (in-store vs. takeaway) allows for channel attribution -- a strategically critical distinction in post-pandemic food retail.

SECTION 3

Problem Statement & Objectives

Formal Problem Definition

Research Question: How do item performance, payment preferences, and time-based sales trends impact overall cafe revenue, and which factors drive peak sales periods?

Project Scope

This analysis is bounded to the calendar year 2023, covering all 12 months of transactional activity. The dataset encompasses 8 menu items, 3 payment methods, and 2 service locations. The analytical scope excludes employee-level performance, cost-side data, and customer-level cohort analysis, as these fields are not present in the available dataset.

Objectives

#	Objective	Analytical Method
O1	Identify top-performing items by revenue and volume	Item pivot: SUM Total Spent, SUM Quantity
O2	Determine payment method revenue contribution	Payment method pivot: SUM Total Spent
O3	Map monthly revenue trends and peak periods	Monthly pivot: SUM, COUNT, AVERAGE
O4	Analyse in-store vs. takeaway channel split	Location pivot: SUM Total Spent
O5	Identify cross-dimensional revenue patterns	Cross-tabulation pivots (item x location, etc.)

Success Criteria

- All five objectives addressed with quantitative evidence from the dataset.
- A minimum of 8 actionable business insights derived from cross-dimensional analysis.
- Recommendations mapped directly to specific data findings with impact rationale.
- 100% financial accuracy across all reported revenue figures.

SECTION 4
Data Description

Dataset Source

Provider: Kaggle Open Datasets

[Kaggle Dataset URL]

Dataset Name: Cafe Sales -- Dirty Data for Cleaning Training

Access: Publicly available, no authentication required

[Google Sheets Dashboard]

Dataset Profile

Attribute	Raw Dataset	Cleaned Dataset
Total Records	~10,000 transactions	6,658 validated transactions
Columns	8 fields	9 fields (+ Transaction Month)
File Format	CSV (UTF-8)	CSV (UTF-8)
Temporal Scope	January - December 2023	January - December 2023
Null Values	1,247 instances	0 instances
Duplicate Records	87 exact matches	0 duplicates
Financial Accuracy	77% (23% miscalculated)	100% verified
Data Quality Status	Raw / Uncleaned	Production-Ready

Column Schema

Column	Type	Description	Example Values
transaction_id	String (ID)	Unique transaction identifier	TXN_1961373
item	Categorical	Menu item purchased	Coffee, Salad, Cake
quantity	Numeric (Int)	Units purchased per transaction	1, 2, 3, 4, 5
unit_price	Numeric ($)	Price per unit in USD	$1.00 - $5.00
total_spent	Numeric ($)	Transaction revenue (qty x price)	$4.00 - $25.00
payment_method	Categorical	Payment channel used	Cash, Credit Card, Digital Wallet
location	Categorical	Service channel	In-store, Takeaway
transaction_date	Date (YYYY-MM)	Transaction period	2023-01 - 2023-12
transaction_month	Categorical	Human-readable month label	January - December

Data Limitations

- **Single-year scope:** The 2023 dataset prevents year-over-year comparison or multi-year trend identification.
- **No timestamp granularity:** Data is at month level only; intra-day or hourly peak analysis is not possible.
- **No customer identifiers:** Repeat customer behaviour, loyalty patterns, and cohort analysis cannot be performed.
- **No cost data:** All revenue figures represent top-line sales only; gross margin analysis is not feasible.
- **33.42% data loss:** Removal of 3,342 records may introduce slight selection bias if corruption was non-random.

SECTION 5

Data Cleaning & Preparation

All primary cleaning and transformation steps were executed in Google Sheets, as per the capstone project requirement. The pipeline comprised seven sequential stages, each designed to address a specific category of data quality failure without contaminating records already validated by earlier stages.

Stage	Objective	Records Affected	Outcome
1. Column Standardisation	SQL/BI naming compatibility	All columns	lowercase_snake_case applied
2. Text Normalisation	Eliminate phantom categories	37 phantom categories	Whitespace + casing unified
3. Placeholder Removal	Remove system artifacts	1,089 removed; 1,158 imputed	Zero placeholder values remain
4. Numeric Validation	Enforce mathematical validity	312 type issues; 487 imputed	All fields numeric & positive
5. Financial Recalculation	Guarantee revenue accuracy	2,301 records corrected ($2,847)	100% accuracy verified
6. Logical Filtering	Remove impossible transactions	1,164 records removed	All qty > 0, price > 0
7. Duplicate Detection	Prevent revenue inflation	87 duplicates ($1,243 inflation)	Zero duplicate transaction IDs

Key Transformations Detail

Financial Recalculation was the most critical intervention. A verification column (quantity x unit_price) was compared against the raw total_spent field for every record. The 2,301 discrepancies -- representing a $2,847 error -- were corrected by replacing total_spent with the calculated value. Post-correction, COUNTIF validation confirmed zero residual errors.

Placeholder and Null Handling used a tiered strategy: records with missing transaction IDs or item names were deleted outright, while missing payment_method and location values were restored via distribution-proportional imputation, preserving the dataset's categorical balance without introducing artificial bias.

Feature Engineering: A transaction_month column was derived from transaction_date using Google Sheets text functions to enable human-readable temporal aggregation in pivot tables. This is the only new field added to the schema.

Quality Dimension	Pre-Cleaning	Post-Cleaning	Standard Met
Null Values	1,247 instances	0 instances	Passed
Financial Accuracy	77% (2,301 errors)	100% (0 errors)	Verified
Duplicate Records	87 transactions	0 transactions	Eliminated
Logical Constraints	1,164 violations	0 violations	Enforced
Categorical Integrity	37 phantom categories	Clean taxonomy	Standardised
Schema Compliance	Mixed types & formats	Uniform schema	Validated

SECTION 6

KPI & Metric Framework

Fourteen strategic KPIs were defined across four analytical dimensions. Each metric is directly traceable to one or more project objectives and is computed from verified cleaned data.

KPI	Formula	Dimension	Business Relevance
Total Revenue	SUM(total_spent)	Revenue	Primary performance benchmark
Avg. Transaction Value (ATV)	AVG(total_spent)	Revenue	Basket size & pricing power
Revenue per Item	SUM(total_spent) GROUP BY item	Product	Menu profitability ranking
Item Sales Volume	SUM(quantity) GROUP BY item	Product	Demand intensity by SKU
Top 5 Revenue Generators	Pareto of item revenue	Product	80/20 revenue concentration
Payment Method Revenue	SUM(total_spent) GROUP BY payment	Payment	Channel financial contribution
Avg. Spend by Payment Type	AVG(total_spent) GROUP BY payment	Payment	Basket size by payment channel
Monthly Revenue	SUM(total_spent) GROUP BY month	Temporal	Seasonal demand mapping
Monthly Transaction Count	COUNT(transaction_id) BY month	Temporal	Volume vs. value separation
Monthly Avg. Transaction	AVG(total_spent) GROUP BY month	Temporal	Per-visit spending trend
In-Store vs. Takeaway Split	SUM(total_spent) BY location	Operational	Channel revenue attribution
Item Revenue by Location	SUM(total_spent) BY item, location	Operational	Channel-level menu performance
Monthly Revenue by Location	SUM(total_spent) BY month, location	Operational	Seasonal channel dynamics
Item Revenue by Payment	SUM(total_spent) BY item, payment	Cross-dim.	Purchase behaviour patterns

SECTION 7

Exploratory Data Analysis

7.1 Revenue Summary KPIs

$60,483 Total Verified Revenue (2023)	**$9.09** Average Transaction Value	**6,658** Validated Transactions
8 Menu Items Analysed	**3** Payment Channels	**2** Service Locations

7.2 Item Sales Performance

The item performance analysis reveals a highly concentrated revenue structure. Salad alone accounts for 26.4% of total revenue -- nearly as much as the bottom four items combined. This Pareto imbalance signals a significant dependency risk and a clear menu engineering priority.

Item	Total Revenue ($)	Revenue Share (%)	Units Sold	Avg. Unit Price ($)
Salad	$15,963.50	26.4%	3,732	$4.27
Smoothie	$9,432.00	15.6%	2,358	$4.00
Sandwich	$9,204.00	15.2%	2,301	$4.00
Juice	$7,731.00	12.8%	2,538	$3.05
Cake	$7,376.00	12.2%	2,435	$3.03
Coffee	$4,720.00	7.8%	2,278	$2.08
Tea	$3,570.00	5.9%	2,215	$1.61
Cookie	$2,487.00	4.1%	2,250	$1.12
TOTAL	$60,483.50	100%	20,107	--

Table 7.1 -- Item-level revenue and volume breakdown. Items ranked by total revenue.

A critical insight emerges when comparing revenue share to volume share: Tea and Cookie generate near-equivalent unit volumes to Coffee, yet contribute dramatically less revenue due to lower price points. This indicates that volume-driving low-price items are not efficiently converting foot traffic into revenue. Conversely, Salad's combination of highest price point ($4.27) and highest volume (3,732 units) makes it the standout performer across both dimensions.

7.3 Monthly Revenue Trends

The monthly analysis reveals a revenue band between $4,552.50 (May) and $5,290.50 (January), representing a peak-to-trough variance of $738.00 or 16.2%. This relatively narrow band suggests structurally stable demand rather than dramatic seasonality -- a characteristic of cafe operations with a loyal recurring customer base.

Month	Revenue ($)	Transactions	Avg. Transaction ($)	Quantity Sold
January	$5,290.50	570	$9.28	1,729
February	$4,754.50	505	$9.41	1,561
March	$5,274.00	588	$8.97	1,762
April	$4,859.50	529	$9.19	1,610
May	$4,552.50	511	$8.91	1,519
June	$5,276.00	568	$9.29	1,729
July	$4,894.00	555	$8.82	1,635
August	$5,107.00	571	$8.94	1,697
September	$5,074.50	566	$8.97	1,720
October	$5,204.00	585	$8.90	1,776
November	$5,171.00	566	$9.14	1,719
December	$5,026.00	543	$9.26	1,650

Table 7.2 -- Monthly revenue, transaction count, average transaction value, and total units sold.

Three distinct revenue clusters emerge: **Peak months** (January, March, June -- all above $5,270) exhibit high transaction counts driven by volume rather than basket size. **Mid-tier months** (August through December -- $5,026-$5,204) show consistent performance. **Trough months** (February, April, May, July -- $4,552-$4,894) record lower transaction counts, suggesting demand-side weakness. February's ATV of $9.41 is the highest of the year, confirming that fewer but higher-value visits characterise quiet periods.

7.4 Payment Method Analysis

Payment method analysis reveals that Cash remains the plurality channel by both transaction count and total revenue, but the gap between Cash and digital methods is narrower than might be expected in a modern retail environment.

Payment Method	Total Revenue ($)	Revenue Share (%)	Transactions	Avg. Transaction ($)
Cash	$23,649.00	39.1%	2,585	$9.15
Digital Wallet	$18,604.50	30.8%	2,052	$9.07
Credit Card	$18,230.00	30.1%	2,020	$9.02
TOTAL	$60,483.50	100%	6,657	$9.09

Table 7.3 -- Revenue and transaction distribution by payment method.

The average transaction value is nearly identical across all three payment methods (range: $9.02-$9.15), which is a statistically significant finding. It indicates that payment method choice does not correlate with basket size. This debunks a common retail assumption that card payments drive higher-value transactions. Payment infrastructure decisions should therefore be evaluated on cost and friction reduction, not on hoped-for basket size uplift.

7.5 Service Channel Analysis (In-Store vs. Takeaway)

Channel	Total Revenue ($)	Revenue Share (%)	Transactions	Avg. Transaction ($)
In-Store	$42,522.00	70.3%	4,647	$9.15
Takeaway	$17,961.50	29.7%	2,010	$8.94
TOTAL	$60,483.50	100%	6,657	$9.09

Table 7.4 -- Revenue and transaction count by service channel.

In-store service generates 70.3% of total revenue -- a clear signal that the cafe's competitive advantage lies in its dine-in experience. The volume gap (4,647 in-store vs. 2,010 takeaway transactions) rather than a spending gap explains the revenue differential, since average transaction values are comparable ($9.15 vs. $8.94).

SECTION 8

Advanced Analysis

8.1 Item Revenue by Location (Cross-Tabulation)

Cross-tabulating item revenue by service channel reveals which products are channel-agnostic versus those with strong dine-in or takeaway preference. This informs menu design decisions for potential takeaway-specific optimisation.

Item	In-Store ($)	Takeaway ($)	Total ($)	In-Store Share (%)
Salad	$11,290.00	$4,673.50	$15,963.50	70.7%
Smoothie	$6,840.00	$2,592.00	$9,432.00	72.5%
Sandwich	$6,288.00	$2,916.00	$9,204.00	68.3%
Juice	$5,550.00	$2,181.00	$7,731.00	71.8%
Cake	$5,146.00	$2,230.00	$7,376.00	69.8%
Coffee	$3,204.00	$1,516.00	$4,720.00	67.9%
Tea	$2,524.00	$1,046.00	$3,570.00	70.7%
Cookie	$1,680.00	$807.00	$2,487.00	67.5%

Table 8.1 -- Item revenue split between in-store and takeaway channels.

Sandwich and Coffee exhibit the lowest in-store revenue share (68.3% and 67.9% respectively), suggesting stronger takeaway affinity relative to other menu items. Smoothie commands the highest in-store share (72.5%), indicating it is primarily a dine-in, occasion-based purchase. This supports a targeted takeaway promotion strategy centred on Sandwich and Coffee bundles.

8.2 Monthly Revenue by Payment Method

Tracking payment method revenue month-by-month reveals an emerging shift in consumer payment preferences over the year -- a critical input for payment infrastructure planning.

Month	Cash ($)	Credit Card ($)	Digital Wallet ($)
January	$2,087.50	$1,719.50	$1,483.50
February	$1,787.50	$1,470.50	$1,496.50
March	$2,197.00	$1,613.50	$1,463.50
April	$1,853.50	$1,427.50	$1,578.50
May	$1,760.50	$1,363.00	$1,429.00
June	$2,129.50	$1,548.00	$1,598.50
July	$1,948.50	$1,418.50	$1,527.00
August	$2,015.00	$1,743.50	$1,348.50
September	$2,056.50	$1,460.50	$1,557.50
October	$2,026.00	$1,407.00	$1,771.00
November	$1,911.50	$1,458.00	$1,801.50
December	$1,876.00	$1,600.50	$1,549.50

Table 8.2 -- Monthly revenue breakdown by payment method.

A notable trend emerges in Q4: Digital Wallet revenue grows from $1,348.50 in August to $1,801.50 in November -- a 33.6% increase in three months. By November, Digital Wallet is within $110 of Cash for that month, signalling a structural shift in consumer payment preference accelerating toward year-end.

8.3 Location Revenue by Payment Method

Location	Cash ($)	Credit Card ($)	Digital Wallet ($)	Total ($)
In-Store	$16,896.50	$12,832.50	$12,793.00	$42,522.00
Takeaway	$6,752.50	$5,397.50	$5,811.50	$17,961.50
TOTAL	$23,649.00	$18,230.00	$18,604.50	$60,483.50

Table 8.3 -- Revenue cross-tabulation: location by payment method.

Within the takeaway channel, Digital Wallet ($5,811.50) already outpaces Credit Card ($5,397.50), suggesting that on-the-go customers prefer friction-reduced mobile payments. This argues for prioritising contactless and digital wallet acceptance infrastructure at the takeaway counter specifically.

SECTION 9

Insights Summary

The following ten insights represent the most decision-relevant findings from the complete analytical framework. Each is stated in business language and supported by quantitative evidence.

I1

Salad Dominates Revenue with Structural Dependency Risk

Salad generates $15,963.50 -- 26.4% of total annual revenue. Its simultaneous leadership in both unit price ($4.27) and volume (3,732 units) creates a revenue concentration risk. Any supply disruption, ingredient cost spike, or competitor substitution poses a disproportionate revenue threat.

I2

Low-Price Items Generate Volume Without Commensurate Revenue

Cookie, Tea, and Coffee collectively account for 33.1% of total units sold (6,743 units) but only 17.8% of revenue ($10,777). These items attract foot traffic but underperform on revenue contribution -- a classic menu engineering challenge.

I3

January and March Drive the Highest Revenue Peaks

January ($5,290.50) and March ($5,274.00) are the top two revenue months. They also record among the highest transaction counts (570 and 588 respectively), confirming that peak months are volume-driven, not spend-per-visit driven.

I4

May Is the Weakest Month -- By Both Volume and Revenue

May records the lowest revenue ($4,552.50) and the fewest transactions (511). The ATV of $8.91 is also below the annual average ($9.09), indicating that May underperforms across all three primary revenue drivers simultaneously.

I5

In-Store Channel Accounts for 70.3% of Revenue

With $42,522.00 generated in-store versus $17,961.50 takeaway, the cafe's primary value driver is clearly the dine-in experience. The per-transaction gap is small ($9.15 vs. $8.94), confirming that takeaway underperformance is a volume problem -- not a spending-level problem.

I6

Payment Method Has No Measurable Impact on Basket Size

Cash ($9.15 ATV), Digital Wallet ($9.07), and Credit Card ($9.02) are statistically equivalent in average spend. Payment method selection should therefore be guided by transaction cost and processing efficiency -- not by hoped-for basket size uplift.

I7

Cash Leads Revenue But Digital Wallet Is Closing the Gap in Q4

Digital Wallet revenue grew 33.6% from August to November ($1,348.50 to $1,801.50). By November, Digital Wallet is within $110 of Cash for that month, signalling an accelerating structural shift in consumer payment preference.

I8	**Sandwich and Coffee Have the Highest Takeaway Affinity** Sandwich and Coffee have the lowest in-store revenue share (68.3% and 67.9% respectively). These are natural candidates for takeaway-specific bundle promotions and packaging investment.

I9	**Revenue Is Remarkably Stable Month-to-Month** The peak-to-trough revenue range of $738.00 (16.2% variance) across 12 months indicates structurally stable demand. The cafe does not experience dramatic seasonal swings, simplifying inventory planning, staffing, and financial forecasting.

I10	**Smoothie Commands Premium In-Store Positioning** Smoothie has the highest in-store revenue share of any item (72.5%), indicating it is almost exclusively consumed in-store. At $4.00 per unit with 2,358 units sold, it should be prominently featured in in-store merchandising and menu positioning.

SECTION 10

Recommendations

Each recommendation below is mapped directly to an analytical insight and includes a business impact rationale and implementation feasibility assessment.

R1 | Priority: High | Based on: I1

Diversify the Revenue Base Away from Salad Dependency

Introduce a premium Salad variant (e.g., seasonal or protein-enhanced) at a 20-25% price premium while simultaneously promoting Smoothie and Sandwich as secondary revenue anchors through combo pricing. Reducing revenue concentration below 22% for any single item improves financial resilience.

Implementation Feasibility: *Low -- requires menu design and supplier negotiation; no capital investment required.*

R2 | Priority: Medium-High | Based on: I2

Engineer Upgrades for Low-Revenue, High-Volume Items

Introduce Add-On pairings for Cookie, Tea, and Coffee (e.g., Cookie + Coffee at a 10% bundle discount). If 20% of the 6,743 units sold across these items convert to a bundle adding $2.00 in revenue, this generates approximately $2,697 in incremental annual revenue.

Implementation Feasibility: *Low -- requires POS system update and staff training only.*

R3 | Priority: High | Based on: I3

Align Staffing and Inventory Peaks with January, March, and June

These three months generate 26.5% of annual revenue ($15,840.50) despite representing 25% of the year. Ensure full inventory availability, reduced stockout risk, and optimised staffing ratios during these periods. A 5% operational efficiency gain in peak months recovers approximately $792 in revenue.

Implementation Feasibility: *Low -- operational scheduling change with no capital requirement.*

R4 | Priority: Medium | Based on: I4

Deploy Targeted Promotions in May and February

May and February are the two lowest-revenue months ($4,552.50 and $4,754.50). A targeted promotion -- such as a loyalty double-points event or a Monday Bundle offer -- targeting a 5% transaction volume increase in these months would generate approximately $500-$600 in additional revenue.

Implementation Feasibility: *Low-Medium -- requires promotional planning and minor marketing spend.*

R5 | Priority: Medium-High | Based on: I7, I8

Invest in Takeaway Digital Wallet Infrastructure

Digital Wallet is already the leading payment method in the takeaway channel ($5,811.50 vs. $5,397.50 for Credit Card) and is growing rapidly in Q4. Ensuring zero-friction digital payment acceptance at the takeaway counter -- including NFC terminals and QR code payment options -- directly supports channel growth trajectory.

Implementation Feasibility: *Medium -- requires terminal hardware investment; one-time capital cost.*

R6 | Priority: Medium | Based on: I10

Position Smoothie as the Flagship In-Store Experience Item

Smoothie's 72.5% in-store revenue share and $9,432.00 annual revenue make it the defining in-store beverage. Feature it prominently in table-top displays, seasonal variants, and loyalty rewards to reinforce the in-store visit occasion and drive repeat dine-in behaviour.

Implementation Feasibility: *Low -- marketing and menu placement only.*

R7 | Priority: Medium | Based on: I6

Standardise Digital Payment Acceptance for Operational Cost Reduction

Since payment method has no measurable effect on basket size, decisions about payment channel prioritisation should be driven entirely by transaction processing cost. Cash handling typically costs more than digital processing at scale. A payment cost audit is recommended to quantify potential savings.

Implementation Feasibility: *Low -- requires payment processor analysis; no operational change required.*

SECTION 11

Impact Estimation

The following estimates are derived from conservative assumptions applied to the verified 2023 revenue baseline of $60,483.50. All estimates assume partial implementation success and do not account for compounding effects.

Recommendation	Mechanism	Est. Revenue Impact	Est. Cost Saving
R1 -- Salad Premium Variant	15% of 3,732 units upgrade to premium (+$1.20/unit)	+$6,717 / year	--
R2 -- Bundle Upsell	20% bundle conversion at +$2.00/unit	+$2,697 / year	--
R3 -- Peak Month Operations	5% efficiency gain in Jan, Mar, Jun	+$792 / year	~$400 reduced overtime
R4 -- Off-Peak Promotions	5% volume lift in May & February	+$548 / year	--
R5 -- Takeaway Digital Wallet	Reduce cash friction; grow takeaway by 3%	+$539 / year	~$800/year cash cost reduction
R6 -- Smoothie Positioning	10% Smoothie volume increase	+$943 / year	--
R7 -- Payment Cost Audit	Shift 10% cash volume to digital	--	~$600/year
TOTAL (Conservative)	--	+$12,236 / year	~$1,800 / year

Table 11.1 -- Conservative impact estimates for all seven recommendations.

In aggregate, the seven recommendations represent a conservative incremental revenue opportunity of $12,236 per annum -- a 20.2% increase on the 2023 baseline -- combined with approximately $1,800 in operational cost savings. The total annual value creation potential of approximately $14,036 from low-to-medium feasibility interventions represents a compelling return on the analytics investment.

SECTION 12

Limitations

Data Limitations

- **Single-year dataset:** With only 2023 data available, it is impossible to distinguish structural trends from one-time anomalies.
- **Month-level granularity only:** The absence of date or time-of-day data prevents intra-day demand analysis, which would be essential for staffing optimisation.
- **No customer identifiers:** Repeat customer frequency, loyalty behaviour, and cohort analysis cannot be computed.
- **No cost data:** All analysis is revenue-side only. Gross margin and profitability analysis are not possible.
- **33.42% record loss:** The removal of 3,342 records may introduce selection bias if corruption was non-randomly distributed.

Assumption Risks

- **Distribution-based imputation:** Missing payment_method and location values were restored using observed distributions, assuming data is Missing at Random (MAR). If missingness correlates with specific months or items, the imputed values may introduce systematic error.
- **Revenue estimates:** All impact estimates in Section 11 are based on simple percentage assumptions. They do not account for price elasticity, competitor response, or capacity constraints.

What Cannot Be Concluded

- Year-over-year growth trajectory or whether 2023 was a recovery, growth, or plateau year.
- Whether observed payment method trends will continue beyond 2023.
- Whether individual customers are driving repeat revenue or whether the customer base is entirely transactional.
- The profitability of any item -- only its revenue contribution is known.

SECTION 13

Future Scope

Additional Analysis Opportunities

- **Hourly demand analysis:** Integrating timestamp data would enable intra-day peak identification, allowing precise staffing schedules and targeted happy-hour promotions.
- **Multi-year trend analysis:** With 2024 data, year-over-year comparisons would validate whether observed seasonal patterns are structural or anomalous.
- **Customer cohort analysis:** Adding customer identifiers would enable RFM (Recency, Frequency, Monetary) segmentation, loyalty programme design, and churn prediction.
- **Margin analysis:** Incorporating cost-of-goods-sold data would shift the analysis from revenue optimisation to profit optimisation.
- **Predictive demand forecasting:** With sufficient historical data, time-series models (ARIMA, Prophet) could forecast monthly demand with confidence intervals.
- **Payment trend forecasting:** The observed Q4 shift toward Digital Wallet payments warrants a dedicated trend analysis to forecast when Digital Wallet may overtake Cash.

New Data Required

- Transaction timestamps (hour/minute) for intra-day analysis
- Customer identifier or loyalty programme ID for cohort analysis
- Cost-of-goods-sold (COGS) per item for margin analysis
- Employee or shift identifiers for labour cost attribution
- Promotional codes or discount flags for marketing attribution
- Table or order number for dine-in experience analysis
- Multi-year historical data (2022-2024) for trend validation

SECTION 14

Conclusion

This report has delivered a comprehensive, evidence-based answer to the research question: *how do item performance, payment preferences, and time-based sales trends impact overall cafe revenue, and which factors drive peak sales periods?*

The analysis demonstrates that cafe revenue is fundamentally driven by three structural realities. First, **item performance is the primary revenue lever:** Salad's dominance at 26.4% of total revenue reflects both its price premium and consumer demand intensity, while low-price high-volume items (Cookie, Tea, Coffee) represent under-monetised foot traffic. Second, **payment method choice is a consumer preference, not a revenue driver:** the near-identical average transaction values across Cash ($9.15), Digital Wallet ($9.07), and Credit Card ($9.02) mean that payment infrastructure decisions should optimise for cost and friction -- not for hoped-for basket size uplift. Third, **temporal demand is structurally stable with modest seasonal variation:** the 16.2% peak-to-trough revenue range confirms consistent consumer behaviour year-round, with January, March, and June as identifiable high-demand periods and May as a reliable trough requiring targeted intervention.

The service channel analysis reinforces that the cafe's competitive moat lies in its in-store dine-in experience (70.3% of revenue), while takeaway represents a structurally equivalent per-transaction channel that is underperforming primarily on volume. The emerging Digital Wallet adoption in the takeaway channel in Q4 provides a clear strategic signal for infrastructure investment.

Seven recommendations -- all rated low-to-medium implementation feasibility -- collectively represent a conservative revenue uplift potential of $12,236 per annum (20.2% on the 2023 baseline), combined with $1,800 in operational cost savings.

Value Delivered: From raw, unstructured POS data to seven actionable business recommendations with a quantified $14,036 annual value creation opportunity -- achieved through rigorous data engineering, systematic analytics, and business-language interpretation of every finding.

SECTION APPENDIX

Appendix

A. Data Dictionary

Field	Type	Allowed Values	Null Policy	Notes
transaction_id	String	TXN_XXXXXXX format	NOT NULL	Primary key; unique per transaction
item	Categorical	Coffee, Tea, Cookie, Cake, Juice, Salad, Sandwich, Smoothie	NOT NULL	8 valid categories
quantity	Integer	> 0	NOT NULL	Imputed via item median where missing
unit_price	Decimal ($)	> 0.00	NOT NULL	Fixed by item; $1.00-$5.00 range
total_spent	Decimal ($)	> 0.00	NOT NULL	Recalculated as qty x unit_price
payment_method	Categorical	Cash, Credit Card, Digital Wallet	NOT NULL	Imputed via distribution where missing
location	Categorical	In-store, Takeaway	NOT NULL	Imputed via distribution where missing
transaction_date	String (YYYY-MM)	2023-01 to 2023-12	NOT NULL	Month-year precision only
transaction_month	Categorical	January-December	NOT NULL	Derived from transaction_date

B. Contribution Matrix

Team Member	Role	Dataset	Cleaning	KPI & Analysis	Dashboard	Report
Shaik Tajuddin	Project Lead & Architect	Yes	Yes	Yes	Yes	Yes
Debashish Karan	Data Engineer	Yes	Yes	--	--	--
Akshat Chauhan	Analytics Specialist	Yes	--	Yes	--	--
Parthraj Singh	Visualisation Lead	Yes	--	Yes	Yes	--
Pankaj Baid	Business Strategist	Yes	Yes	--	--	--
Krishna Verma	QA & Documentation	Yes	--	--	--	Yes

Declaration: We confirm that the above contribution details are accurate and verifiable through Google Sheets version history and submitted project artifacts. All analysis is original work produced for the Newton School of Technology Data Analytics Capstone, Academic Year 2024-2025.